Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2025

Code: 101401 | ***Subject: Civil Engineering-Societal & Global Impact*** | *Time: 03 Hours*

Semester: IV | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which of the following is a traditional method of solid waste management?
- (i) Composting
- (ii) Waste-to-energy
- (iii) Recycling
- (iv) Landfilling

(b) Which of the following methods is used for hazardous waste treatment?
- (i) Incineration
- (ii) All of the above
- (iii) Landfilling
- (iv) Chemical neutralization

(c) What is the main function of dams in flood control?
- (i) Generating electricity
- (ii) Storing water to control downstream flooding
- (iii) Providing water for irrigation
- (iv) Diverting water to reservoirs

(d) How does GIS help in disaster management?
- (i) By predicting weather events
- (ii) By monitoring air pollution
- (iii) By mapping disaster-prone areas and planning evacuation routes
- (iv) By providing real-time traffic data

(e) Which of the following was a negative environmental consequence of the Green Revolution?
- (i) Increased soil fertility
- (ii) Decrease in deforestation
- (iii) Excessive use of chemical fertilizers and pesticides
- (iv) Improved water conservation

(f) By how much has atmospheric carbon dioxide concentration increased ever since the Industrial Revolution?
- (i) 20%
- (ii) 10%
- (iii) 40%
- (iv) 60%

(g) What is the main purpose of a railway signaling system?
- (i) To prevent train accidents
- (ii) To monitor train speed
- (iii) To provide communication between stations
- (iv) To generate electricity

(h) The structural concrete slab panels can be repaired by _____
- (i) Internal bonding
- (ii) External bonding
- (iii) Stress bonding
- (iv) Layered bonding

(i) The preparation of the subgrade does not involve
- (i) Grading
- (ii) Site clearance
- (iii) Compaction
- (iv) Spreading

(j) Bituminous material is used in highway construction primarily because of them
- (i) Cementing and waterproof properties
- (ii) High specific gravity
- (iii) Load-bearing capacity
- (iv) Black colour which facilitates

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) What are multi-purpose water projects, and why are they important?

(b) What are the methods for the conservation, repair, and rehabilitation of structures and heritage buildings?

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) What are intelligent/smart buildings, and how do they enhance sustainability?

(b) What is sound control in built environments, and why is it important?

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Define LEED ratings and EIA. What is hyper loops? How it can play a role in futuristic growth of nation.

(b) Define Ocean thermal energy conversion. Why it is in news now a days?

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Q.5 Solve both questions : **[7 x 2 = 14]**

(a) What are the benefits of using Lean Construction in civil engineering projects?

(b) How can civil engineering projects reduce their carbon footprint?

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Recently which energy is declared as renewable energy by GOI? How it can have impact on the total energy generation of country?

(b) How does the infrastructure development & growth of the nation can have effects on employments and living standards of people?

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) What are the major environmental concerns in modern times?

(b) What role do policies play in environmental management?

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Examine the role of GIS in modern urban planning and environmental monitoring.

(b) Enlist the modern engineering marvels of India. Describe any one of them in detail.

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) Various goals of sustainable development

(b) Environment Protection Acts

(c) Global warming

(d) Green Infrastructure

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